

LPKF Laser & Electronics SE

LPK.DE | Xetra SDAX | Laser Manufacturing

Comprehensive Equity Research Report

LIDE glass technology | Advanced packaging | CPO optionality

€5.77

Share price

~€127M

Market cap

€115.3M

FY25 revenue

~0%

FY25 adj EBIT margin

Investment thesis: LPKF is a 50-year-old German laser manufacturing specialist with a diversified portfolio (PCB prototyping, plastic welding, solar thin-film scribing, electronics manufacturing) that is repositioning around a single disruptive technology: **LIDE (Laser Induced Deep Etching)**, a patented process for creating defect-free through-glass vias (TGVs) at aspect ratios up to 1:50 with sub-micron accuracy. LIDE is emerging as the enabling technology for glass-based advanced semiconductor packaging — a structural shift driven by Intel, Samsung, and TSMC-led roadmaps for glass core substrates and glass interposers in AI/HPC chips. Further, LPKF is developing LIDE-based CPO (co-packaged optics) integration on glass substrates — a second-generation opportunity layered on top. FY2025 revenue was €115.3M (-6.2% YoY) with near-breakeven adjusted EBIT. The 'North Star' transformation programme targets a sustainable double-digit EBIT margin by 2028. **Rating: SPECULATIVE — structural AI/packaging option with execution and timing risk.**

Metric	Value	Metric	Value
Ticker	LPK.DE / LPK.ETR	Founded	1976 (50 years)
Sector	Industrial lasers / semiconductor	Employees	~700
Segments	Development / Electronics / Welding / Solar	HQ	Garbsen (Hanover), Germany
Index	SDAX (Prime Standard)	CEO / CFO	Dr. Klaus Fiedler / Peter Mümmeler
FY25 revenue	€115.3M (-6.2% YoY)	FY24 revenue	€122.9M
FY25 adj EBIT	€0.8M (vs €0.1M FY24)	FY25 adj EBIT margin	~0%
FY25 order intake	€91.6M (-19.9%)	FY25 order backlog	€27.0M (-€23.9M YoY)
Key technology	LIDE (Laser Induced Deep Etching)	Flagship system	NEXAR LIDE 5000
Target (2028)	Double-digit EBIT margin	Transformation program	North Star (launched Sept 2025)

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1. Executive summary

LPKF Laser & Electronics SE (LPK.DE) is a Germany-based technology company founded in 1976, headquartered in Garbsen near Hanover, and listed in the SDAX segment of the Frankfurt Stock Exchange. LPKF develops and manufactures laser-based solutions across four segments: Development (PCB prototyping), Electronics (semiconductor packaging + SMT), Welding (plastic component joining), and Solar (thin-film module scribing). The company employs ~700 people globally and generated €115.3M revenue in FY2025 [1][2][3].

The investment case pivots on a single transformational technology: **LIDE (Laser Induced Deep Etching)**, a patented two-step process combining ultrafast laser modification with selective wet etching to create defect-free through-glass vias (TGVs) at aspect ratios up to 1:50 with sub-micron accuracy. LIDE addresses the critical bottleneck in glass-based advanced semiconductor packaging — a structural shift endorsed by Intel (glass core substrates roadmap), Samsung, and TSMC for next-generation AI/HPC chips [4][5][6].

In parallel, LPKF is developing a second-generation opportunity: **CPO (co-packaged optics) integration on glass substrates**. Glass's unique combination of electrical insulation, dimensional stability (<2µm flatness at 600×600mm panel scale), CTE match with silicon, and microfluidic cooling integration makes it architecturally superior to organic substrates for AI-era optical interconnects. LPKF has initiated cooperation programmes to build a glass-focused CPO supply chain [6][7].

FY2025 delivered €115.3M revenue (-6.2% YoY) with adjusted EBIT of €0.8M (margin ~0%) — a slight improvement over FY2024 despite weak demand. Order intake fell to €91.6M (-19.9%) reflecting the absence of major solar orders. Management launched the **North Star** transformation programme in September 2025 targeting a sustainable double-digit EBIT margin by 2028, funded by an extended syndicated loan agreement through 2028 [3][8]. **Rating: SPECULATIVE** — the LIDE opportunity is real and structural, but timing of the advanced packaging ramp remains uncertain and dependent on downstream qualification cycles at customers.

2. Company overview & history

LPKF was founded in 1976 by civil engineer Jürgen Seebach in Hanover, Germany. The company name derives from the original product: 'LeiterPlattenKopierFräse' — a Circuit Board Copy Milling machine for in-house PCB prototyping. This foundation product remains a core revenue contributor 50 years later. The company went public in 1998 and is listed on the Prime Standard segment of Deutsche Börse (SDAX) [1][9].

LPKF's product evolution followed adjacent-technology expansion: from milling machines to laser-based PCB prototyping (ProtoLaser series), then laser plastic welding systems for automotive and medical devices, then laser scribing for thin-film solar cell manufacturing, and most recently laser-based semiconductor glass processing via LIDE. Today, LPKF operates through four segments, with the semiconductor/advanced packaging opportunity representing the company's strategic future focus under the 2025-launched North Star transformation [1][8].

Key management: **Dr. Klaus Fiedler** (CEO, contract through Dec 2028) and **Peter Mümmeler** (CFO, contract through March 2028). LPKF operates globally with subsidiaries across Europe, North America, and Asia, and has production sites in Germany (Garbsen, main), Suhl, and Fürth. The ARRALYZE life sciences division was partially relocated to Boston in 2024-25 [1][2][9].

3. Business segments overview

Segment	Description	Status / outlook	Strategic priority
Electronics	Laser systems for electronics production and glass component manufacturing — includes LIDE and SMT/stencil cutting	Key growth segment; semiconductor/advanced packaging ramp	Highest strategic priority — focus of North Star programme
Development	PCB prototyping systems for R&D; and in-house circuit board production	Solid growth prospects per management	Established profitable core; steady cash generation
Welding	Laser beam welding systems for plastic components (automotive, medical, consumer electronics)	Recently realigned; expected to return to profitable long-term growth	Stable profitable segment post-realignment

Segment	Description	Status / outlook	Strategic priority
Solar	Laser scribing for thin-film solar cells; laser transfer printing for functional inks	Currently weak — major order drought; future upside in perovskite transition	Lowest near-term priority; perovskite is a long-term option

Figure 1: LPKF segment overview. Sources: [1][2][3][8]

Segment dynamics FY2025: Electronics (semiconductor-adjacent) and Development (PCB prototyping) are positioned as growth drivers. Welding has stabilised post-restructuring. Solar is currently the biggest drag — the absence of major thin-film solar orders was the primary cause of the €91.6M FY2025 order intake (-19.9% YoY). Management hopes the industry transition to perovskite solar cells will re-ignite this segment, but near-term visibility is poor [3].

4. Technology thesis: why glass in advanced packaging?

The semiconductor industry is migrating from traditional organic laminate substrates to glass core substrates and glass interposers for the highest-performance AI/HPC chips. This transition is driven by structural limitations of existing materials as chips scale to multi-die chiplet architectures, HBM stacks, and next-generation high-bandwidth interconnects [4][5][10].

Attribute	Organic laminate (legacy)	Silicon interposer	Glass substrate (new)
Dielectric constant	~3-4	~12 (lossy at high freq)	~4-6
CTE (ppm/°C)	~15-18 (large mismatch)	~3 (silicon match)	3.2-3.5 (silicon match)
Panel size	Up to 510×515mm	300mm wafer only	Up to 600×600mm with <2µm flatness
Dimensional stability	Warp at scale	Good but wafer-limited	Excellent — enables chiplet integration
Thermal / cooling	No integrated cooling	No integrated cooling	Microfluidic channel integration possible
Cost at scale	Low (baseline)	High (wafer economics)	Mid — panel-level processing lowers per-unit cost
High-frequency signal	Adequate to ~10GHz	Lossy at multi-GHz	Multi-GHz suitable for HBM, chiplets, CPO

Figure 2: Substrate material comparison. Sources: [4][6][10]

Industry validation: Intel announced glass core substrate roadmap for 2026-2030. Samsung and SK Hynix are investing in glass-based HBM stacking. TSMC's advanced packaging roadmap includes glass interposer options. Market research firms project panel-level glass packaging as a >\$5B market by 2030, with growth accelerating from <\$100M in 2025 as first volume ramps begin [5][10][11].

Where LPKF fits: The critical bottleneck in glass-based packaging is creating through-glass vias (TGVs) — precise holes through the glass that are then metallised to carry electrical signals. Conventional approaches (mechanical drilling, chemical etching, traditional laser ablation) all create micro-cracks or substrate splitting ('SeWaRe' effect), leading to yield failures. LPKF's LIDE process is one of the few proven crack-free TGV technologies at production throughput [6][12].

5. LIDE: LPKF's core differentiator

LIDE (Laser Induced Deep Etching) is LPKF's patented two-step process. **Step 1 — laser modification:** Ultrafast laser pulses 'mark' the glass structure without removing material, creating chemically-modified zones with sub-micron accuracy. **Step 2 — selective wet etching:** A rapid batch etching bath selectively removes only the laser-modified zones, leaving pristine shapes (vias, channels, 2.5D features) that maintain the glass's inherent mechanical strength. The result: no heat-affected zones, no micro-cracks, no subsurface damage [12][13].

Parameter	Specification	Significance
Aspect ratio	Up to 1:50	Enables deep vias in thin glass — critical for HBM / chiplet stacks
Via accuracy	Sub-micron	Matches the alignment precision of advanced logic nodes
Sidewall quality	Zero micro-cracks, zero chipping	Unique among laser-based TGV methods — critical for yield
Flagship system	NEXAR LIDE 5000	Production-scale throughput; panel-level processing
Complementary systems	NEXAR Ablate 5000 (RDL removal), NEXAR Bond 5000 (glass-glass bonding)	LPKF building broader process portfolio beyond TGV only
Bonding throughput	Up to 200 mm/s (Tensor Bonding)	Direct glass-glass bonds without adhesives — optical transparency preserved
Service model	Equipment sales + Vittron foundry (small-to-high volume manufacturing service)	Dual model: sells machines to IDMs; offers contract manufacturing for fabless customers

Figure 3: LIDE technology specifications. Sources: [6][12][13]

Vertical integration strategy: Historically LPKF offered only the TGV creation step. In FY2025, management announced expansion along the value chain to include: (1) **singulation** (dicing) of glass-based packages; (2) **laser-based bonding** of multilayer glass stacks (Tensor Bonding); (3) **ablation** for RDL (redistribution layer) removal. This broadens LPKF's role from a single-process tool vendor to a multi-process strategic partner — potentially multiplying revenue per customer 3-5x [8][12].

6. CPO on glass: the next-generation opportunity

As detailed in our prior SIVE.ST and Largan (3008.TW) analyses, Co-Packaged Optics (CPO) is reshaping AI data center architecture. Traditional pluggable optical transceivers (at the front panel of a switch, 15-30cm from the ASIC) are being replaced by optical engines co-packaged directly alongside the processor silicon. NVIDIA's Spectrum-X, Broadcom's Bailly, and TSMC's COUPE platform are commercialising CPO for mass production in H2 2026 onwards [14][15].

Why glass is the natural CPO substrate:

- **Optical transparency** — glass can carry light waveguides directly integrated into the package, eliminating lossy electrical-to-optical conversion steps [6][14]
- **Thermal management** — glass uniquely combines electrical insulation with microfluidic coolant channels that can be routed alongside high-speed signals without interference — impossible with conductive silicon substrates
- **V-groove precision** — LIDE creates precision V-grooves for positioning optical fibre arrays, a critical requirement for CPO's fibre-to-chip coupling
- **CTE match** — glass matches silicon photonic ICs (3.2-3.5 ppm/°C), eliminating thermal cycling stress that limits reliability
- **Panel-level economics** — 600×600mm panel processing dramatically lowers per-package cost vs. 300mm silicon interposers

LPKF's CPO positioning: LPKF explicitly stated in its FY2025 annual report that it is '*working on the integration of co-packaged optics (CPO) on glass substrates — a forward-looking application with high revenue potential*'. The company is building partnerships across the glass-CPO supply chain. Unlike SIVE.ST (supplies the CW laser device) or IQE (supplies the photonic epi wafer), LPKF's role is at the **packaging layer** — providing the manufacturing equipment and process IP to integrate photonic ICs with glass substrates [4][6].

CPO value chain layer	Key players	LPKF's role
Laser light source (CW)	Lumentum, Coherent, SIVE, MACOM	Not active
Photonic IC (PIC) fab	TSMC, GlobalFoundries, IMEC	Not active
PIC epi wafers	IQE, Sumitomo, II-VI	Not active
Fibre array unit (FAU)	FOCI, Himax (for TSMC COUPE)	Not active
Prism-based collimators	Largan (via PhotoniCore)	Not active
Glass substrate / interposer	Corning, AGC, Nippon Electric Glass	Process equipment + Vitrion foundry service
TGV processing	LPKF (LIDE), Samtec, others	CORE BUSINESS — process equipment IP leader
Multi-layer glass bonding	Emerging field	Tensor Bonding technology (NEXAR Bond 5000)
Assembly / packaging	ASE, Amkor, SPIL	Not active (but customers of LPKF equipment)

Figure 4: CPO value chain and LPKF's positioning. Sources: [14][15][16]

7. Customer ecosystem & partnerships

LPKF does not disclose specific semiconductor customer names — a feature of the B2B equipment industry, where NDAs with chipmakers are standard. However, the following pattern emerges from public disclosures, trade press, and partnership announcements [1][4][6]:

Customer type	Examples / pattern	Status
Semiconductor IDMs and foundries	Glass TGV equipment for advanced packaging R&D; and early volume (Intel, TSMC, Samsung-class customers)	Evaluation and pilot orders; mass production ramps await downstream qualification
Glass substrate suppliers	Corning, AGC, Nippon Electric Glass supply bare glass; LPKF partners for process integration	Cooperation programme announced Sept 2024 to build glass supply chain
OSATs (outsourced assembly/test)	ASE, Amkor, SPIL — eventual large buyers of LIDE systems	Follow-on demand once IDM qualification complete
Display industry	Smartphone, AR/VR display glass microstructuring	Established revenue stream via Vitrion foundry
PCB prototyping (Development)	Industrial customers, research institutes globally — thousands of ProtoLaser systems installed	Stable cash cow; recurring consumables revenue
Plastic welding (Welding seg.)	Automotive Tier-1 suppliers, medical device OEMs	Recurring equipment refresh cycle
Solar manufacturers	Thin-film (CIGS, perovskite) cell makers	Currently weak order book
Life sciences (ARRALYZE)	Boston-based expansion for precision glass component production	Emerging application

Figure 5: Customer ecosystem. Sources: [1][4][6][17]

8. Competitive landscape

Competitor	Country	Revenue	Focus	vs. LPKF
TGV equipment competitors	Various	n/a	Samtec, Philoptics, ZIC lasers, other laser TGV specialists	Smaller scale; less integrated value chain
Corning (GLW)	USA	~\$13B	Glass supplier — potential forward integration threat into TGV	Massively larger; could in-source if market grows
Applied Materials	USA	~\$27B	Broad equipment portfolio; could acquire or develop TGV capability	500x larger; strategic threat if glass packaging becomes critical
ASML / Onto Innovation	Netherlands / USA	Large	Packaging metrology and lithography — complementary rather than direct competition	Ecosystem partners, not direct competitors
PCB prototyping competitors	Various	n/a	Bungard Elektronik, T-Tech, MITS Electronics	LPKF is global leader in PCB prototyping
Laser plastic welding	Various	n/a	TRUMPF, Rofin-Sinar (now Coherent), Emerson	LPKF specialises in premium plastic welding applications
Thin-film solar scribing	Various	n/a	ROFIN, TRUMPF, Han's Laser (China)	Chinese competition intense; market structurally weak
LPKF	Germany	€115.3M	LIDE + laser plastic welding + PCB prototyping + solar scribing	Unique LIDE IP position in emerging glass packaging market

Figure 6: Competitive landscape. Sources: [1][2][18]

The competitive dynamic is two-sided: in established segments (PCB prototyping, plastic welding, solar scribing), LPKF faces commoditisation pressure from Asian competitors (Han's Laser in particular). In the emerging LIDE / glass packaging segment, LPKF is one of the earliest movers with deepest IP, but faces potential forward integration from massive-scale ecosystem players (Corning, Applied Materials) if the market becomes strategically important [2][18].

9. Financial analysis: 3-year review

Metric (€M)	FY2022	FY2023	FY2024	FY2025
Revenue	125.5	123.6	122.9	115.3
Revenue growth	+20%	-1.5%	-0.6%	-6.2%
Adjusted EBIT	~6.0	~1.5	~0.1	~0.8
Adjusted EBIT margin	~5%	~1%	~0%	~1%
Reported net income	~2.0	~ -1	~ -5	~ -3
Order intake	~135	~120	~114	91.6
Order intake growth	+15%	-11%	-5%	-19.9%
Order backlog (year-end)	~40	~38	~51	27.0
Capex	~10	~11	~8	~6
Cash position	~15	~12	~10	~10-12

Figure 7: 3-year financial summary. FY2025 actuals from LPKF annual report. Earlier years partly estimated from press releases. Sources: [3][19]

Observations: (1) Revenue has declined for three consecutive years (€125.5M → €115.3M), reflecting weak end-market demand across segments. (2) Adjusted EBIT has stabilised near zero — management arrested the decline but has not returned to meaningful profitability. (3) Order intake fell sharply in FY2025 (-19.9%), driven by the solar segment's order drought — year-end backlog shrank by €23.9M. (4) LPKF has maintained cash discipline (capex scaled to ~€6M) and extended its syndicated loan facility through 2028 as part of the North Star programme. (5) The company is clearly in a transition/transformation phase — the critical question is whether LIDE revenue materialises before the cost base becomes unsustainable [3][8][19].

10. North Star transformation programme

North Star pillar	Detail
Headline target	Sustainable double-digit EBIT margin by 2028 (currently ~0-1%)
Launched	September 2025
Scope	Five key areas: operations, R&D, sales, administration & infrastructure, service
Mechanism	Organisational restructuring, selected personnel and cost adjustments, process efficiency improvements
Financing	Syndicated loan agreement extended through 2028 — solid financing runway
Advisor	Specialised Hanover-based consulting firm supporting the leadership team
Strategic focus	Semiconductor/advanced packaging expansion (LIDE) as primary growth lever; profitable maintenance of core segments (Development, Welding)
FY2026 expectation	Cautious outlook; solar segment decline and meaningful transformation costs expected; no significant growth expected short-term
Potential upside	High-volume orders from advanced packaging NOT included in FY2026 forecast — timing determined by downstream customer qualification, not LPKF process readiness

Figure 8: North Star programme details. Sources: [3][8]

Critical investor point: Management explicitly states that 'potential high-volume orders from the advanced packaging business are not included' in the FY2026 forecast because the ramp timing is 'determined not by the LIDE process provided by LPKF, but by the qualification of downstream process steps' at customers. This means the LIDE upside is entirely **off-baseline** — any advanced packaging ramp would be incremental to guidance. It also means investors should not expect 2026 to deliver the thesis; 2027-2028 is the realistic inflection window [3][8].

11. Key ratios & valuation

Ratio	Value	Assessment
P/E (TTM)	N/A	Loss-making; no meaningful P/E
EV/Sales (TTM)	~1.1x (€127M mcap, minimal net debt)	Modest vs. semi equipment peers (3-5x)

Ratio	Value	Assessment
EV/EBIT	High (near-zero EBIT)	Not meaningful at current profitability
P/B	~1.5x	Reasonable for industrial tech
Gross margin	~40% (historical average)	Decent; reflects equipment + service mix
EBIT margin (FY25)	~0.7%	Near-breakeven — target 10%+ by 2028
Adj net debt	Minimal	Strong balance sheet; financed through 2028
Dividend	Suspended	Capital preserved for transformation
Analyst consensus	Mixed (Warburg Buy)	Limited coverage at small-cap German tech level
52-wk range	~€4.65 - €8.50 (approx.)	Volatile; trading closer to range midpoint

Figure 9: Valuation metrics. Sources: [1][2][19]

Valuation framing: At €5.77 and ~€127M market capitalisation, LPKF trades at ~1.1x FY25 revenue. This is a **deep discount** to semi equipment peers (ASML, Applied Materials, KLA trade at 3-10x revenue on profitable businesses). The discount reflects (a) near-zero current profitability, (b) revenue decline trajectory, (c) uncertainty over LIDE timing. A successful transformation — returning to €140-160M revenue with 10% EBIT margin by 2028 — would justify 2-3x revenue multiple, implying €10-15 per share range (+70-160% from current). Failure would imply continued grind at €5-6 [19][20].

12. Growth catalysts & path to profitability

Timing	Catalyst	Potential impact
Ongoing (2026)	North Star efficiency measures execute	Cost base reduction; EBIT margin progress from ~1% toward mid-single digits
H2 2026	First meaningful LIDE orders for advanced packaging volume production	€5-10M incremental revenue; proof point for thesis
2026-2027	Intel, Samsung, TSMC glass substrate programme ramps	Binary on individual customer wins; €10-30M revenue potential per major win
2027	CPO-on-glass pilot deployments	Early revenue; validates second-generation opportunity
2027-2028	NEXAR LIDE 5000, Ablate 5000, Bond 5000 adopted as multi-process portfolio	Revenue per customer 3-5x higher than TGV-only approach
2028	Double-digit EBIT margin target year	If achieved, 2-3x revenue multiple re-rating; €10-15 per share
2028+	Perovskite solar ramp	Potential re-emergence of Solar segment as growth contributor
Ongoing	Strategic partnership announcements / cooperation programme expansions	Incremental credibility; potential LOIs from major customers

Figure 10: Growth catalyst timeline. Sources: [3][6][8][20]

13. Peter Lynch framework

Criterion	Assessment	Signal
Category	Turnaround / asset play with embedded technology option	Neutral
Revenue CAGR (3Y)	-2.8% (declining)	Negative
PEG ratio	N/A	Negative
Gross margin	~40%	Positive (equipment + service mix)
EBIT margin	~0-1%	Negative — must improve for thesis to work
Balance sheet	Minimal net debt; syndicated loan through 2028	Positive (solid financing)
Analyst coverage	1-3 analysts (under-followed)	Positive (contrarian)
Earnings history	Briefly profitable at peak; current losses	Neutral-negative
The 'story'	LIDE / glass packaging / CPO secular technology	Positive
Complicated operations?	Four segments, diverse end markets	Negative — diversification dilutes focus
Insider alignment	Key management contracted through 2028 (CEO, CFO)	Positive (stability for transformation)

Figure 11: Peter Lynch framework applied to LPKF

Under Lynch's framework, LPKF is a **turnaround with embedded technology optionality**. Lynch was cautious on turnarounds but believed in paying for optionality when underlying technology was genuine. The LIDE glass-packaging thesis aligns with Lynch's 'fast grower within a slow grower' pattern — an emerging business line that could transform the company, embedded within a mature industrial laser franchise [21].

14. Risk factors & bear case

Risk factor	Severity	Description
Advanced packaging timing risk	CRITICAL	Management explicitly states timing is determined by downstream customer qualification, not LPKF readiness. The ramp could be 2026, 2027, 2028, or later — with corresponding impact on cash runway
Competitive displacement	HIGH	Corning, Applied Materials could forward-integrate; alternative TGV processes (mechanical, chemical, plasma) could win qualification
Solar segment drag	HIGH	Solar order drought accounts for bulk of FY25 order intake decline; perovskite transition is multi-year
North Star execution	MEDIUM-HIGH	Transformation programmes often slip or under-deliver; organisational restructuring risks talent departure
Small-cap liquidity / coverage	MEDIUM-HIGH	SDAX listing with limited analyst coverage = sharp price moves on limited volume
Cyclical pressure on core segments	MEDIUM	PCB prototyping and plastic welding tied to industrial capex cycles
China competition (solar, welding)	MEDIUM	Han's Laser and Chinese equipment makers pressure margins in established segments
Customer concentration (glass)	MEDIUM	LIDE's major customers likely concentrated in a handful of IDMs — slow qualification at any major customer delays overall ramp
Exchange rate	LOW-MED	German export business; EUR appreciation hurts USD-denominated revenue
Management succession	LOW	CEO contract through 2028, CFO through early 2028 — stability for transformation period

Figure 12: Risk factor assessment

15. Conclusion

Factor	Bull case	Bear case
Technology	LIDE is proprietary, patented, and addresses a structural industry need — glass in advanced packaging is 'when not if'	Alternative TGV processes catch up; Corning or Applied Materials in-source; LIDE becomes commoditised
Revenue	LIDE volume orders begin H2 2026; €140-180M revenue by 2028	LIDE ramp slips to 2028+; revenue stuck at €110-130M
Profitability	North Star achieves double-digit EBIT by 2028; operating leverage on LIDE ramp	North Star delivers 4-5% EBIT; no material LIDE contribution; fixed costs weigh
Balance sheet	Syndicated loan through 2028 provides runway without dilution	Cash burn accelerates; equity raise needed at low multiple
Strategic option	CPO-on-glass adds second-generation upside in 2028+	CPO on glass loses to alternative substrate architectures
Valuation	Re-rate to 2-3x revenue at €140-160M = €10-15/share (+70-160%)	Continued €5-6 range; deep-value trap

Figure 13: Bull/bear assessment summary

Rating: SPECULATIVE. LPKF is a genuine emerging-technology option embedded in a struggling industrial laser business. The LIDE / glass advanced packaging / CPO thesis is structurally sound — validated by Intel, Samsung, TSMC roadmaps and by LPKF's expansion into adjacent process steps (Ablate, Bond). The ~€127M market capitalisation on €115M revenue offers modest absolute downside versus meaningful upside if LIDE ramps. However, management explicitly excludes high-volume advanced packaging orders from the FY2026 forecast — meaning the thesis hinges on customer qualification cycles outside LPKF's control.

Unlike SIVE.ST (high-margin photonics with near-term Ayar Labs catalyst) or IQE (distressed strategic review with takeover premium), LPKF is a patient-capital thesis. Investors need to believe in (a) LIDE's technical superiority, (b) the glass-packaging transition, and (c) LPKF's ability to retain its IP lead as Corning, Applied Materials, or Asian competitors enter. Multi-year horizon (2027-2028 inflection) and high risk tolerance required.

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